

基于混合检索器的多模态生成模型研究与实现

中期答辩

时子延

19220422

指导老师: 宋歌

计算机与电子信息学院/人工智能学院

南京师范大学

2026 年 3 月 23 日



1. 研究概述

1.1. 研究目的

1.2. 意义与现状

2. 研究内容与方法

2.1. 研究内容

2.2. 技术路线

3. 研究进度与结果

3.1. 研究进度

3.2. 阶段结果

4. 后续计划与预期效果

4.1. 实验计划

4.2. 预期效果

5. 支撑材料

5.1. 论文与过程材料

5.2. 实验与现场展示

6. 参考文献

研究目标

- ▶ 面向多模态检索增强生成 (mRAG) 场景，提升检索器返回有效视觉证据的能力。
- ▶ 基于 MRAG-Bench 构建统一评测流程，回答“更懂指令的检索器是否真的能提升下游问答效果”。
- ▶ 以 MagicLens 为核心检索器，对比 GT Oracle、Random、Text-only 与 corpus 检索等策略。

挑战

- ▶ 多模态问答中，性能瓶颈究竟在“候选从哪里来”，还是“候选如何排序”。
- ▶ 当问题真正依赖视觉证据时，现有检索器是否能稳定召回并利用正确图像。

研究动机

- ▶ LLM/MLLM 仍存在知识冻结与幻觉问题，RAG 通过外部知识缓解真实性问题。
- ▶ 在多模态场景下，错误不仅是“文本事实编造”，还包括“视觉事实编造”。
- ▶ 因此需要从文本 RAG 走向 mRAG，并重点提升视觉证据检索质量。

研究意义

- ▶ 为具身智能提供可检索的长期记忆与证据链。
- ▶ 为脑机接口与高可靠辅助沟通提供更低幻觉的检索基础。
- ▶ 为后续检索器训练与扩散模型驱动增强提供实验基础。

mRAG 研究主线

- 框架层：mRAG 从静态单轮走向动态、按需、多轮检索，核心关注幻觉抑制与时效性提升。
- 表示层：CLIP、ALIGN 等统一语义空间方法推动跨模态检索成为可能。
- 检索层：FG-CLIP、Long-CLIP、MagicLens 等工作分别从细粒度理解、长文本建模、开放式指令检索改进效果。

目前仍存在的空缺

- 现有研究更关注通用框架综述，缺少面向“视觉证据是否真正被找对”的系统评测。
- 多数实验没有拆开粗召回与重排两类能力，难以定位 mRAG 的真实瓶颈。

Published as a conference paper at ICLR 2025

MRAG-BENCH: VISION-CENTRIC EVALUATION FOR RETRIEVAL-AUGMENTED MULTIMODAL MODELS

Wenbo Hu¹, Jia-Chen Gu¹, Zi-Yi Dou¹, Mohsen Fayyaz¹, Pan Lu²,

Kai-Wei Chang¹, Nanyun Peng¹

¹UCLA, ²Stanford University

{wenbohu, gujc, zdou}@ucla.edu

<https://mragebench.github.io>

ABSTRACT

Existing multimodal retrieval benchmarks primarily focus on evaluating whether models can retrieve and utilize external *textual knowledge* for question answering. However, there are scenarios where retrieving visual information is either more beneficial or easier to access than textual data. In this paper, we introduce a multimodal retrieval-augmented generation benchmark, MRAG-BENCH, in which we systematically identify and categorize scenarios where visually augmented knowledge is better than textual knowledge, for instance, more images from varying viewpoints. MRAG-BENCH consists of 16,130 images and 1,353 human-annotated multiple-choice questions across 9 distinct scenarios. With MRAG-BENCH, we conduct an evaluation of 10 open-source and 4 proprietary large vision-language models (LVLMs). Our results show that all LVLMs exhibit greater improvements when augmented with images compared to textual knowledge, confirming that MRAG-BENCH is vision-centric. Additionally, we conduct extensive analysis with MRAG-BENCH, which offers valuable insights into retrieval-augmented LVLMs. Notably, the top-performing model, GPT-4o, faces challenges in effectively leveraging retrieved knowledge, achieving only a 5.82% improvement with ground-truth information, in contrast to a 33.16% improvement observed in human participants. These findings highlight the importance of MRAG-BENCH in encouraging the community to enhance LVLMs' ability to utilize retrieved visual knowledge more effectively.

出发点

- ▶ MRAG-Bench 是 vision-centric benchmark。
- ▶ 核心问题不是“能否检索文本知识”，而是“视觉证据是否比文本更关键”。
- ▶ 这与本文从图像检索器切入 mRAG 的研究目标直接一致。

Published as a conference paper at ICLR 2025

Statistic	Number
Total questions	1,353
- Multiple-choice questions	1,353 (100%)
- Questions newly annotated	1,353 (100%)
Total Scenarios	9
Unique number of questions	375
Unique number of answers	463
Total number of images	16,130
Unique number of images	16,130
Human selected images	9,673
Average image size (px)	1076 x 851
Maximum question length	20
Maximum answer length	9
Average question length	8.03
Average answer length	2.16
Average choice number	4

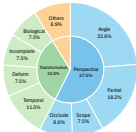


Table 2: Key statistics of MRAG-BENCH. Figure 2: Scenarios distribution of MRAG-BENCH.

Notably, we observe while all models improve with GT knowledge, only proprietary models are able to effectively utilize noisy retrieved multimodal knowledge. This indicates the gap between open-source and close-source models still exists. Open-source models are falling short on their parametric knowledge and the ability to distinguish between high-quality and poor-quality retrieved visually augmented examples. In comparison to humans, GPT-4o achieves only a 5.82% improvement when augmented with GT knowledge and 0.28% with retrieved knowledge, whereas humans demonstrate a 33.16% and 22.91% improvement, respectively. These results highlight the importance of MRAG-BENCH in encouraging the community to develop LVLMs better utilizing of visually augmented knowledge.

2 MRAG-BENCH

2.1 BENCHMARK OVERVIEW

Our benchmark is designed for systematic evaluation of LVLm’s vision-centric multimodal RAG abilities. To achieve this, we focus on evaluating the model’s understanding of image objects that are not commonly associated with its knowledge base, while the collected ground-truth images can help incentivize specific visual concepts within LVLm’s memory. Therefore, we divide our benchmark into two main aspects, as illustrated in the examples in Figure 1:

- *perspective*, refers to the challenges in visual recognition and reasoning that arise when a visual entity is presented from varying viewpoints, scopes, or levels of visibility.
- *transformative*, refers to the challenges that arise when a visual entity undergoes fine-grained physical transformations, making it unfamiliar or not easily associated with the model’s prior knowledge.

Published as a conference paper at ICLR 2025



Figure 3: Qualitative examples on MRAG-BENCH. For each scenario, we show the result of GPT-4o (OpenAI, 2023), Gemini Pro (Team et al., 2023), LLaVA-Next-Interleave (Li et al., 2024b) and Mantis-8B-Siglip (Jiang et al., 2024a). The ground-truth answer is in blue.

2.2 BENCHMARK COMPOSITION

MRAG-BENCH provides a systematic evaluation across 9 distinctive multimodal RAG scenarios, with four scenarios focused on the perspective understanding of visual entities, four on transformative understanding, and one categorized as “others”. As illustrated in Figure 2, each scenario comprises 7.5% to 23.8% of the whole benchmark. The selected examples of each scenario is shown in Figure 3. The details of each scenario are introduced as follows.

- 1. 研究概述
 - 1.1. 研究目的
 - 1.2. 意义与现状
- 2. 研究内容与方法
 - 2.1. 研究内容
 - 2.2. 技术路线
- 3. 研究进度与结果
 - 3.1. 研究进度
 - 3.2. 阶段结果
- 4. 后续计划与预期效果
 - 4.1. 实验计划
 - 4.2. 预期效果
- 5. 支撑材料
 - 5.1. 论文与过程材料
 - 5.2. 实验与现场展示
- 6. 参考文献

研究内容

1. 基于 MRAG-Bench 重构视觉证据导向的 mRAG 评测任务，将检索阶段作为核心变量。
2. 引入 MagicLens 作为多模态检索器，研究其在“关系驱动证据检索”中的实际价值。
3. 设计 GT Oracle、Random、Retrieved-Rerank、Corpus-CLIP 等对照实验，量化不同检索策略的收益与上限。

组织思路

- ▶ 从整体准确率、分场景准确率、GT 覆盖率与失败模式四个层面分析检索器表现。
- ▶ 将实验明确拆为 **Rerank line** 和 **Corpus line**。
- ▶ 分别回答排序收益与粗召回收益。



为什么引入 MagicLens

- ▶ 传统图像检索更偏向视觉相似性，难表达“内部视角”“相关物品”“不同时刻”等开放关系。
- ▶ MagicLens 将检索定义为 $(image_q, instruction) \rightarrow image_t$ ，强调关系驱动的目标搜索。
- ▶ 这与 MRAG-Bench 中大量依赖视觉证据与跨图关系的题型高度契合。

论文要点

- ▶ 开放式 instruction 让检索从“像不像”转向“关系是否匹配”。
- ▶ MagicLens 在多类检索 benchmark 上表现稳定，同时保持较好的参数效率。

arXiv:2403.19651v2 [cs.CV] 24 Jun 2024

MagicLens: Self-Supervised Image Retrieval with Open-Ended Instructions

Kai Zhang^{1*} Yi Liao^{2*} Huiyang He² Kenton Lee² Siyuan Qiao² Wenbo Chen² Yu Su¹ Ming-Wei Chang²

Abstract

Image retrieval, i.e., finding desired images given a reference image, inherently encompasses rich, multi-faceted search intents that are difficult to capture solely using image-based measures. Recent works leverage text instructions to allow users to more freely express their search intents. However, they primarily focus on image pairs that are visually similar and/or can be characterized by a small set of pre-defined relations. The core thesis of this paper is that text instructions can enable retrieving images with richer relations beyond visual similarity. To show this, we introduce MagicLens, a series of self-supervised image retrieval models that support open-ended instructions. MagicLens is built on a key novel insight: image pairs that naturally occur on the same web page contain a wide range of implicit relations (e.g., *location view of*), and we can bring these implicit relations explicit by synthesizing instructions via foundation models. Trained on 36.7M (query image, instruction, target image) triples with rich semantic relations mined from the web, MagicLens achieves results comparable with or better than prior best on eight benchmarks of various image retrieval tasks, while maintaining high parameter efficiency with a significantly smaller model size. Additional human analyses on a 1-OM image dataset corpus further demonstrate the diversity of search intents supported by MagicLens. Code and models are publicly available at the Project Website.

1. Introduction

Image retrieval is a long-established problem in computer vision (Efthys et al., 2009; Gordo et al., 2016) with a wide

^{*}Work done at Google DeepMind. ¹The Ohio State University ²Google DeepMind. Correspondence to: Kai Zhang (kzhang1525@osu.edu).

Proceedings of the 41st International Conference on Machine Learning, Vienna, Austria. PMLR 235, 2024. Copyright 2024 by the author(s).



Figure 1. Top-1 retrieved images using MagicLens and the prior state-of-the-art (SOTA) method (Gu et al., 2023) from a retrieval pool with 1-OM images. The prior SOTA method, while accepting text instructions, primarily retrieves images based on visual similarity to the query image, ignoring the nuances of the text instructions. In contrast, MagicLens excels at retrieving both visually similar images and those that align with the deeper meaning and context of the text instructions — even when the images do not resemble the query. For example, if given a query image of the Burj Al Arab and the instruction “Find other attractions in this country”, it can successfully locate images of the Palm Jumeirah in Dubai.

range of real-world applications, such as visual search, object localization, and re-identification. However, since its inception, this task has suffered from ambiguous definitions due to the complex and rich content encapsulated in images. Similar images may differ in key aspects, and different images can share commonalities. In image search scenarios, users frequently present multiple search intents for a single query image, indicating that more image relevance is insufficient for precise search results. For instance, when searching with an image of the Burj Al Arab in Dubai (see Figure 1), a user might seek other attractions in Dubai or an interior view, each relating differently to the query image. Therefore, incorporating text instructions that articulate search intents is essential and indispensable for enhancing retrieval accuracy. Ideally, models should accurately capture and interpret diverse real-world search intents as conveyed by open-ended text instructions.

These open-ended search instructions, upon a wide range of topics and concepts, reflect the diverse ways users



自监督数据构造

- ▶ 从同一网页中挖掘自然共现图像对，而不是只依赖人工定义关系。
- ▶ 用 LMM/LLM 为图像对生成开放式 instruction，形成 (query image, instruction, target image) 三元组。
- ▶ 最终构建约 3670 万条训练样本，覆盖大量隐式语义关系。

模型结构

- ▶ 共享参数的 dual-encoder，以 CLIP/CoCa 为初始化。
- ▶ 通过 self-attention 融合图像与指令，再用 attention pooling 压缩成单向量表示。
- ▶ target 端使用 ($image_t$, "") 编码，并采用 contrastive loss；同时把 query image 本身作为 hard negative。

MagicLens: Self-Supervised Image Retrieval with Open-Ended Instructions

PaLI (Chen et al., 2023a) to generate captions. Each type of metadata provides textual information about images from different perspectives. See Appendix A for more details.

(f) *Scoring & Filtering*. After obtaining groups of images along with their extensive metadata, we pair up images within the same group and eliminate unbalanced pairs using a combination of relevance measures. We use the CLIP image-to-image score to assess visual relevance and the text-to-text score for non-visual relevance. Image pairs that don't meet the criteria, such as those with low scores in both aspects, are excluded from consideration. To avoid the over-sampling of substandard images and duplicated relations, we set a maximum of three pairs for each group, thereby ensuring a more uniform distribution of images and relations in our training data (see Figure 5).

Open-Ended Instruction Generation. With the informative metadata of high-quality paired images, LLMs are able to well understand the image content (ICA and captions) and their background information (Alt-text). Using instruction (Chang et al., 2023), few-shot demonstrations (Brown et al., 2020), and chain-of-thought prompting (Wei et al., 2022) techniques, PaLM2 (Anil et al., 2023) generates open-ended instructions that precisely connect the paired images ($image_q, image_t$). Figure 3 illustrates generated instructions and Appendix 10 shows the detailed prompt and demonstrations. Eventually, we obtain 36.7M examples ($image_q, inst, image_t$) for self-supervised image retrieval training.

3.2 MagicLens Model

Model Design. As shown in Figure 4, we adopt a simple dual-encoder architecture with shared parameters and initialize the backbone vision and language encoders with CoCa (Yu et al., 2022) or CLIP (Radford et al., 2021). To enable deep modality integration, we introduce multiple self-attention layers and design a single multi-head attention pooler, compressing the multimodal inputs into a single embedding e for later matching. Additionally, since the retrieval target comprises only an image without accompanying text, we employ an empty text entry "" to transform the target into a multimodal input. We denote e_q as the embedding for the multimodal query ($image_q, inst$) and e_t as the embedding for the target ($image_t, ""$). Considering the efficiency, we propose MagicLens-B and MagicLens-L, initialized with the base and large checkpoints, respectively.

Model Training. We use a simple contrastive loss to train MagicLens. Our model is updated by contrasting the paired query-target against other targets in one training batch. In particular, as the query image itself ($image_q$) can be a challenging hard negative for the multimodal query ($image_q, inst$), we combine the query image itself and an empty text to encode ($image_q, ""$) to get e_q' as an additional query neg-

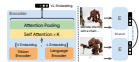


Figure 4. Model architecture and training of MagicLens Encoder (E), which takes the vision and language embeddings and fuses them as a sequence to self-attention layers for modality integration.

ative example. To scale up the number of negative examples, for each query image, we use all query negatives and other target negatives in the same batch. Formally, for the i -th training example, the loss function $\mathcal{L}_i$ is defined as,

$$-\log \frac{\exp(\frac{e_q \cdot e_t}{\tau})}{\exp(\frac{e_q \cdot e_t}{\tau}) + \sum_{j \neq t} \exp(\frac{e_q \cdot e_j}{\tau}) + \sum_{j \neq q} \exp(\frac{e_j \cdot e_t}{\tau})}$$

where $\text{sim}(\cdot)$ indicates a cosine similarity function $\frac{e_q \cdot e_t}{\tau}$. N refers to the sampled batch size, and τ is a temperature hyperparameter for logit scaling. Please refer to Appendix A for more implementation details.

4. Experiments

4.1. Experiment Setup

Benchmarks and Metrics. To comprehensively evaluate MagicLens' multimodality-to-image retrieval ability, we consider three related tasks in a zero-shot, one-checkpoint setting: (1) compound image retrieval (CIR), (2) domain transfer retrieval, and (3) conditional image similarity. Each task has different yet limited sets of image relations. Table 2 shows the detailed statistics of five benchmarks.

Compound Image Retrieval. We consider one domain-specific and two open-domain benchmarks to evaluate the model's domain adaptability and its capability on real-world natural images, respectively. FTQ (Wu et al., 2023) is a fashion-domain benchmark with three disjoint retrieval sub-tasks: dress, shirt, and topset. Following previous work (Saito et al., 2023; Baldrati et al., 2023; Gu et al., 2024), we evaluate on the validation set and report recall averaged over sub-tasks. C110 (Lin et al., 2021) is the first dataset constructed on natural images (Suh et al., 2019) with nine pre-defined relations between the query and the target images. It also includes a subset retrieval setting where models retrieve a target image from a dedicated small subset for each query. However, in addition to the limited size of the retrieval pool, it also suffers from false negative issues, as pointed out by Baldrati et al. (2023). We utilize recall (R and R_{sub}) to evaluate standard retrieval and subset retrieval. In contrast, to better align with the real-world large-scale



统一评测流程

1. 输入 query image + question
2. 检索 Top- k 候选图像
3. 将问题与候选图送入 LLaVA
4. 输出多项选择答案并统计指标

对照设置

- ▶ E0: LLaVA + GT-RAG baseline
- ▶ E1/E2: 固定候选集上的 MagicLens rerank
- ▶ E3: CLIP 从完整 corpus 召回 Top-5
- ▶ E6: CLIP 粗召回 + MagicLens 重排
- ▶ E7: MagicLens 直接从完整 corpus 召回

评价指标

整体 Accuracy、九类场景分布、GT 覆盖率、接近 Oracle 的程度，以及“检索对但生成错 / 检索错但生成对”的失败模式。



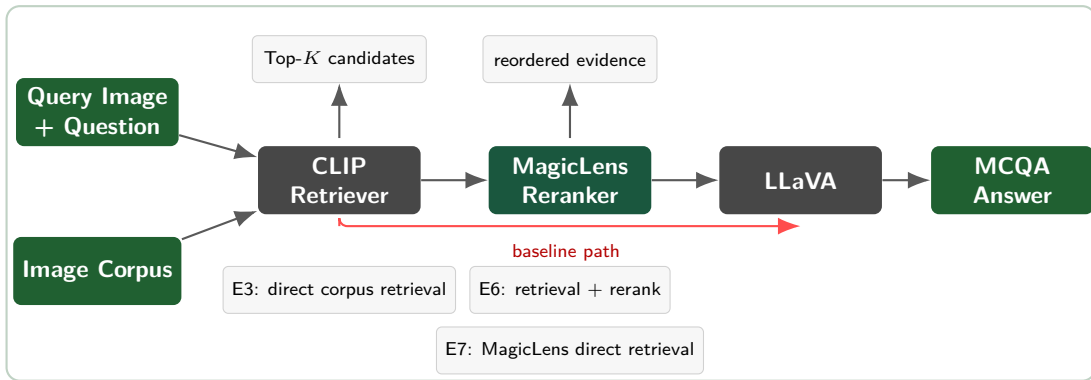
MagicLens 检索 instruction

```
411 def retrieve_corpus_images_magiclens(  
412     query_image,  
413     question_text: str,  
414     corpus_paths,  
415     corpus_embeds,  
416     encode_fn,  
417     tokenizer_fn,  
418     top_k: int,  
419 ):  
420     q_img = preprocess_pil_image(query_image, 224)  
421     q_tok = np.array(tokenizer_fn(question_text))  
422     q_embed = np.asarray(encode_fn(jnp.array(q_tok), jnp.array(q_img)))[0])  
423     q_embed = l2_normalize(q_embed[None, :])[0]  
424     sims = corpus_embeds @ q_embed  
425     k = min(max(1, int(top_k)), len(corpus_paths))  
426     top_idx = np.argpartition(-sims, k - 1)[k:]  
427     top_idx = top_idx[np.argsort(-sims[top_idx])]  
428     ranked = []  
429     for rank, idx in enumerate(top_idx.tolist(), start=1):  
430         path = corpus_paths[idx]  
431         with Image.open(path) as img:  
432             ranked.append(  
433                 {  
434                     "rank": rank,  
435                     "path": str(path),  
436                     "score": float(sims[idx]),  
437                     "image": img.convert("RGB"),  
438                 }  
439             )  
440     return ranked
```

LLaVA 回答 instruction

```
210 def llava_answer(tokenizer, model, image_processor, item, image_files, args):  
211     question_part = item.get("prompt_question_part", item["question"])  
212     if len(image_files) <= 1:  
213         instruction = "Answer with the option's letter from the given choices directly. "  
214     else:  
215         instruction = (  
216             "You will be given one question concerning several images. "  
217             "The first image is the input image, others are retrieved examples to help you. "  
218             "Answer with the option's letter from the given choices directly. "  
219         )  
220     image_tokens = DEFAULT_IMAGE_TOKEN * max(1, len(image_files))  
221     user_query = f"{instruction}{image_tokens}\n{question_part}"
```

回答阶段再把问题、候选图像与 answer-only instruction 一起送入 LLaVA，观察检索噪声对最终作答的影响。



E3

CLIP 直接从完整 corpus 召回 Top- K 候选。

E6

CLIP 粗召回后，再用 MagicLens 重排。

E7

MagicLens 直接作为全库 retriever。



任务定义

$$\begin{aligned} Q &= (x_q, q) \\ I &= R(Q) = \{i_1, \dots, i_K\} \\ \hat{y} &= M(Q, I) \end{aligned}$$

排序分数

$$s_i = \langle z_q, z_i \rangle$$

排序损失

$$\mathcal{L}_{\text{rank}} = -\log \frac{\exp(s(q, i^+))}{\sum_j \exp(s(q, i_j))}$$

联合优化目标

$$\begin{aligned} \mathcal{L}_{\text{ans}} &= -\log p(\hat{y} = y^* \mid Q, I) \\ \mathcal{L} &= \mathcal{L}_{\text{rank}} + \lambda \mathcal{L}_{\text{ans}} \end{aligned}$$

1. 研究概述

1.1. 研究目的

1.2. 意义与现状

2. 研究内容与方法

2.1. 研究内容

2.2. 技术路线

3. 研究进度与结果

3.1. 研究进度

3.2. 阶段结果

4. 后续计划与预期效果

4.1. 实验计划

4.2. 预期效果

5. 支撑材料

5.1. 论文与过程材料

5.2. 实验与现场展示

6. 参考文献



已完成

- ▶ 搭建 MRAG-Bench 数据与评测管线，明确图像候选进入模型的真实规则。
- ▶ 跑通 GT-RAG、Retrieved-Rerank、CLIP corpus Top-5 等基线实验。
- ▶ 完成候选重叠统计与问题拆分，明确区分 rerank 与 corpus 两条实验线。

正在进行

- ▶ 集成 MagicLens 到完整 corpus 检索流程，形成 E6 主实验。
- ▶ 新增 E7 直检索实验脚本，用于评估 MagicLens 作为 direct retriever 的效果。
- ▶ 分场景分析检索收益，定位 Temporal、Scope、Incomplete 等难点场景。



Track	Method	Overall	Perspective				Transformative				Others
			Angle	Partial	Scope	Occlusion	Temporal	Deformation	Incomplete	Biological	
Control	E0: LLaVA+GT-RAG Baseline	59.05	59.63	64.23	66.67	67.59	57.72	56.86	29.41	55.88	64.17
Rerank	E1: <i>GT-RAG-RANKING_{magiclens}</i>	60.16	61.49	65.45	66.67	69.44	58.39	57.84	30.39	55.88	65.00
Rerank	E2: Retrieved-RAG-RANKING _{magiclens}	51.74	51.24	50.41	59.80	59.26	48.99	55.88	33.33	50.00	59.17
Corpus	E3: CLIP-RAG Baseline (Top-5 from corpus)	50.41	52.48	54.07	48.04	54.63	46.31	58.82	22.55	50.00	57.50
Control	E4: LLaVA+No-RAG	53.14	56.21	56.10	55.88	57.41	47.65	55.88	33.33	50.98	55.83
Control	E5: <i>GT-RAG-RANKING_{magiclens}</i> w/o rerank	60.16	59.94	66.67	63.73	66.67	61.74	56.86	29.41	57.84	67.50
Corpus	E6: <i>CLIP-RAG-RANKING_{magiclens}</i>	50.33	54.04	49.19	55.88	51.85	46.31	56.86	27.45	49.02	56.67
Corpus	E7: MagicLens-RAG Baseline (Top-5 from corpus)	47.97	51.55	47.56	45.10	48.15	46.98	50.00	32.35	50.98	51.67



MRAG-Bench 数据特征

- ▶ 共 1353 个测试样本，覆盖 9 类视觉证据依赖场景。
- ▶ GT 候选平均 4.70 张，retrieved_images 原始字段固定为 5 张。
- ▶ Incomplete 场景在 dataloader 中会被截断为 1 张，需单独分析。

重叠统计结论

- ▶ 44.12% 样本中，retrieved 与 GT 完全无重叠。
- ▶ 平均 GT 覆盖率仅 22.78%。
- ▶ 说明数据集自带候选与 oracle 候选存在显著偏差。

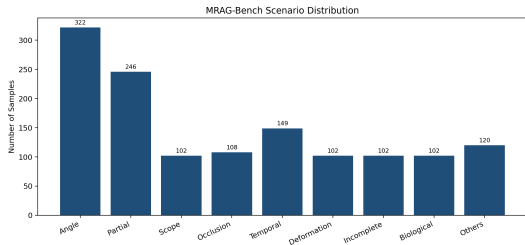
Source	1 张	5 张	Avg.
GT raw	102	1251	4.70
Retrieved raw	0	1353	5.00
Retrieved effective	102	1251	4.70

研究进度与结果

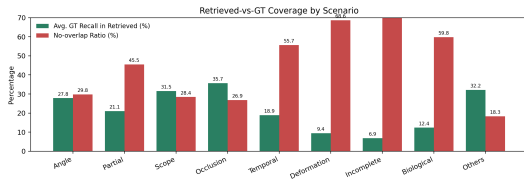
研究结果一：数据集细分分析



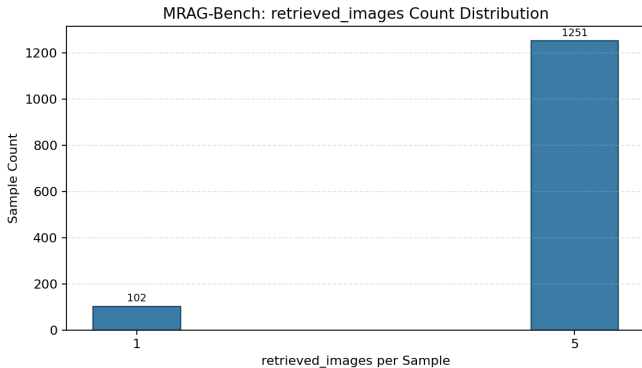
南京师范大学



场景分布并不均衡，Angle、Partial、Temporal、Others 样本量相对更多。



不同场景上的 GT 覆盖率差异明显，说明局部 rerank 与真实全库检索不能混为一谈。



观察

- ▶ 数据集给定候选与 GT 候选分布并不一致。
- ▶ 一部分样本中，retrieved 候选几乎不覆盖真正有效证据。
- ▶ 这也是本文把局部 rerank 与全库检索分开讨论的直接原因。



实验线含义

- ▶ E0: GT-RAG, 上界参考。
- ▶ E1: GT 候选 + MagicLens rerank, 测排序收益。
- ▶ E2: 数据集候选 + MagicLens rerank, 测局部重排能力。
- ▶ E3: CLIP 全库直检索, 测粗召回 baseline。
- ▶ E4: No-RAG, 测无外部证据基线。
- ▶ E5: GT 候选但关闭 rerank, 用于隔离 E1 的排序收益。
- ▶ E6: CLIP 粗召回 + MagicLens rerank, 测混合检索。
- ▶ E7: MagicLens 全库直检索, 测 direct

指标在回答什么

- ▶ Overall Accuracy: 最终问答效果。
- ▶ 分场景 Accuracy: 不同视觉证据类型下的稳定性。
- ▶ E1 vs E5: 判断 rerank 是否真正带来收益。
- ▶ E3 vs E6: 判断重排能否改善 CLIP 粗召回。
- ▶ E3 vs E7: 判断 MagicLens 是否适合直接做全库召回。
- ▶ E4 vs 其余实验: 判断“给证据”是否真的优于“不给证据”。

研究进度与结果

研究结果二：阶段性实验表现



南京师范大学

Overall 对照

Track	Method	Overall (%)
Control	E0: GT-RAG	59.05
Rerank	E1: GT+ML rerank	60.16
Rerank	E2: Ret+ML rerank	51.74
Corpus	E3: CLIP corpus	50.41
Control	E4: No-RAG	53.14
Control	E5: GT no rerank	60.16
Corpus	E6: CLIP+ML rerank	50.33
Corpus	E7: ML corpus	47.97

初步观察

- ▶ E1 与 E5 接近, GT 条件下 rerank 收益有限。
- ▶ E4 高于 E3/E6/E7, 说明噪声证据可能伤害回答。
- ▶ E6 与 E3 接近, 重排尚未稳定改善粗召回。
- ▶ E7 低于 E3, MagicLens 直检索仍弱于 CLIP。

分场景现象

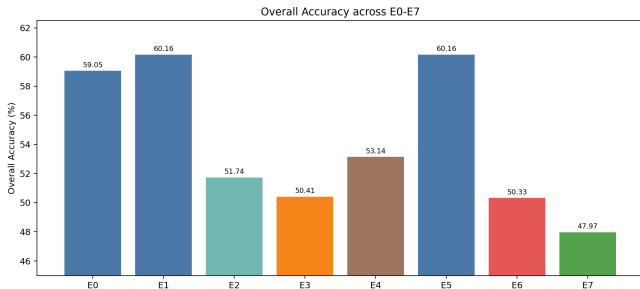
E4 在 Incomplete 上达到 33.33%, 明显高于 E3 的 22.55%; E6 在 Scope 和 Incomplete 上高于 E3, 但在 Partial、Occlusion 与 Biological 上下降, 说明当前提升并不稳定。

研究进度与结果

研究结果二：不同实验整体对比



南京师范大学



整体观察

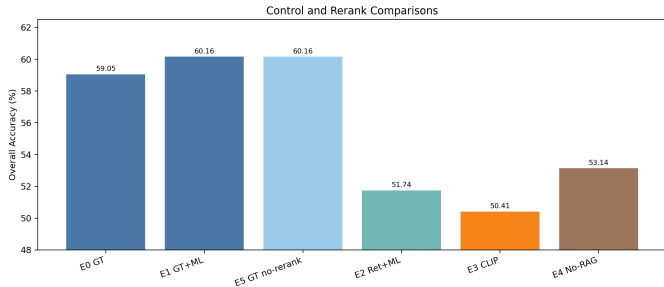
- ▶ E0/E1/E5 代表理想候选条件下的较高上界。
- ▶ E4 高于 E3/E6/E7，说明噪声证据会拉低最终回答效果。
- ▶ E7 目前最低，说明 MagicLens 直检索仍需继续优化 query instruction。

研究进度与结果

研究结果二：控制实验与重排对比



南京师范大学



解读

- ▶ E1 与 E5 基本持平，说明在 GT 条件下 rerank 不是主要瓶颈。
- ▶ E4 高于 E3，说明噪声证据会伤害生成。
- ▶ E2 明显低于 E1/E5，说明数据集候选覆盖不足会直接限制上限。



E7 核心结果

- ▶ 任务：MagicLens 直接从完整 corpus 召回 Top-5。
- ▶ Overall Accuracy: 47.97% (649/1353)。
- ▶ 对比 E3: 低于 CLIP corpus baseline 的 50.41%。

Scenario	E3	E7
Ang.	52.48	51.55
Part.	54.07	47.56
Scope	48.04	45.10
Occl.	54.63	48.15
Temp.	46.31	46.98
Def.	58.82	50.00
Incomp.	22.55	32.35
Bio.	50.00	50.98
Other	57.50	51.67

初步判断

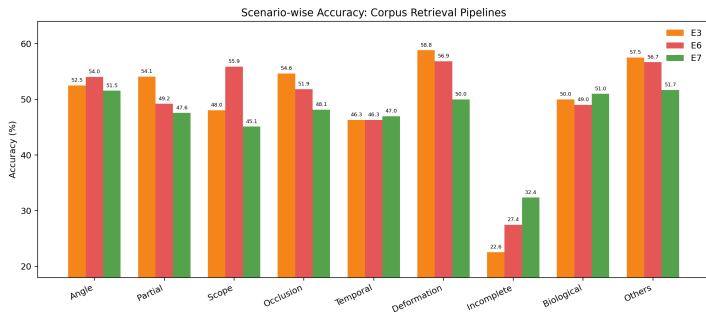
E7 说明 MagicLens 可以直接承担全库召回，但当前整体稳定性仍不如 CLIP；不过它在 Incomplete 场景上有明显优势，说明其在结构缺失类问题上存在潜力。

研究进度与结果

研究结果三：Corpus 线分场景柱状图



南京师范大学



分场景分析

- ▶ E6 在 Scope、Incomplete 上略优于 E3。
- ▶ E7 在 Incomplete 上最强，但在 Partial、Occlusion、Deformation 上明显落后。
- ▶ 当前最稳的 corpus baseline 仍是 E3，说明粗召回质量依旧关键。



Control / Rerank

- ▶ E0: LLaVA + GT-RAG baseline, 作为较高上界。
- ▶ E1: 在 GT 候选集上做 MagicLens rerank。
- ▶ E2: 在数据集给定候选上做 MagicLens rerank。
- ▶ E4: LLaVA + No-RAG, 对应无外部知识基线。
- ▶ E5: GT 候选输入, 但关闭 MagicLens 重排, 用于隔离 rerank 收益。

Corpus 线

- ▶ E3: CLIP 从完整 corpus 直接召回 Top-5。
- ▶ E6: CLIP 粗召回 Top-5, 再由 MagicLens 执行 rerank。
- ▶ E7: MagicLens 直接从完整 corpus 中召回 Top-5, 不经过 CLIP 粗召回。

设计目的

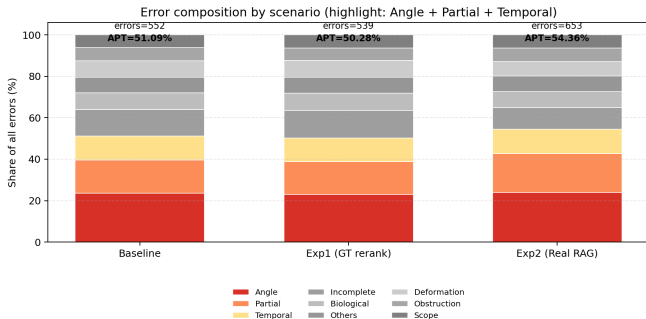
通过 E3、E6、E7 的对照, 区分“CLIP 粗召回”“CLIP+MagicLens 混合检索”与“MagicLens 直接召回”三类能力。

研究进度与结果

研究结果四：错误分布对比



南京师范大学



当前启发

- ▶ 错误并非均匀分布，而是集中出现在若干困难场景。
- ▶ 难点主要落在 Temporal、Scope、Incomplete 等需要强视觉证据的场景。
- ▶ 后续应优先围绕这些场景优化 query 设计、召回策略和重排机制。



阶段结论

- ▶ 仅比较 overall accuracy 不足以解释 mRAG 流程，必须把“候选生成”和“候选排序”分开讨论。
- ▶ MagicLens 在固定候选集上的收益说明其具备一定的排序价值，但这还不等于完整检索能力已经解决。

核心判断

- ▶ 数据集给定候选与 GT 的低重叠率表明，真实 mRAG 的关键问题仍是如何从完整图像库中召回“有用证据”。
- ▶ 从 E1/E5、E3/E6、E3/E7 三组对照看，当前主要矛盾不在“是否加 MagicLens”。
- ▶ 更关键的是给 MagicLens 什么候选、什么 instruction，以及它更适合做 rerank 还是 direct retriever。

- 1. 研究概述
 - 1.1. 研究目的
 - 1.2. 意义与现状
- 2. 研究内容与方法
 - 2.1. 研究内容
 - 2.2. 技术路线
- 3. 研究进度与结果
 - 3.1. 研究进度
 - 3.2. 阶段结果
- 4. 后续计划与预期效果
 - 4.1. 实验计划
 - 4.2. 预期效果
- 5. 支撑材料
 - 5.1. 论文与过程材料
 - 5.2. 实验与现场展示
- 6. 参考文献



实验计划

1. 完成 E6: 在完整 corpus 上采用 CLIP 粗召回 + MagicLens 重排, 验证主方案收益。
2. 训练轻量级 prompt 生成模型, 为 MagicLens 自动生成更适合检索的 query / instruction。
3. 以“提升 GT 命中率、减少噪声候选、最终提升回答准确率”为目标, 系统优化检索提示词设计。
4. 对 Temporal、Scope、Incomplete 等困难场景开展消融分析, 定位失败原因。

进一步工作

- ▶ 补充 No-RAG、w/o rerank 等控制实验, 形成完整的证据链。
- ▶ 比较人工 prompt、规则模板与小模型生成 prompt 的差异, 分析提示词与检索质量之间的关系。
- ▶ 探索扩散模型驱动的数据增强与检索器训练。
- ▶ 对比 E6 与 E7, 判断 MagicLens 更适合作为 reranker 还是 direct retriever。
- ▶ 为后续具身/脑机接口应用奠定基础。



预期研究结论

- ▶ 证明多模态检索器相较于纯文本或弱检索策略具有明确收益。
- ▶ 定位完整 mRAG 中的主要性能瓶颈，判断其更偏向粗召回还是重排。
- ▶ 识别哪些类型的问题真正依赖视觉证据，形成面向后续优化的场景画像。

预期成果形式

- ▶ 一套可复现的 mRAG 检索评测流程与实验协议。
- ▶ MRAG-Bench 上系统性的检索器对比结果与错误分析。
- ▶ 形成一套面向 MagicLens 的提示词优化方法，为提升 GT 覆盖率、抑制噪声检索和改进最终回答效果提供实验基础。

- 1. 研究概述
 - 1.1. 研究目的
 - 1.2. 意义与现状
- 2. 研究内容与方法
 - 2.1. 研究内容
 - 2.2. 技术路线
- 3. 研究进度与结果
 - 3.1. 研究进度
 - 3.2. 阶段结果
- 4. 后续计划与预期效果
 - 4.1. 实验计划
 - 4.2. 预期效果
- 5. 支撑材料
 - 5.1. 论文与过程材料
 - 5.2. 实验与现场展示
- 6. 参考文献

论文目录

1. 研究背景与意义
2. 国内外研究现状
3. 主要研究内容与方法
4. 实验设计与初步结果
5. 结论与后续工作

已形成材料

- ▶ 论文初稿：首页与正文结构已经整理完成。
- ▶ 英文翻译：完成 MRAG-Bench 核心内容翻译。
- ▶ 周记：持续记录阅读、实验和问题排查过程

论文初稿首页

基于混合检索器的多模态生成模型研究与实现

时子延

2026年3月23日

摘要

本文以 MRAG-Bench 为出发点，围绕多模态检索增强生成场景中的检索器选型开展研究。与传统以文本知识检索为核心的 RAG 任务不同，MRAG-Bench 强调视觉证据在特定问题中的决定性作用，因此非常适合用于研究“检索器是否真正找到了有用图像证据”。在此基础上，本文首先复现并分析 MRAG-Bench 评测流程，进一步将问题拆分为局部重排与全局检索两条实验线；随后引入 MagicLens 作为候选重排与混合检索器的核心组件，分析其在关系驱动视觉检索中的潜在价值；最后基于统一评测系统给出当前阶段的初步实验结果，并讨论后续改进方向。

1 研究背景与意义

近年来，以 Transformer 为代表的大规模语言模型与多模态大模型在自然语言理解、视觉问答和跨模态生成等任务中取得了显著进展 [1][2]。然而，模型参数中固化的知识具有明显的时效性与覆盖边界，当问题依赖实时信息、长尾知识或细粒度视觉证据时，仅依靠参数记忆容易出现知识冻结与幻觉问题 [3][4]。检索增强生成 (RAG) 通过在推理阶段引入外部知识，使模型从“闭卷作答”转向“开卷作答”，从而显著提升回答的真实性与可追溯性 [5][6]。

随着多模态大模型的发展，研究重点进一步从文本 RAG 扩展到多模态检索增强生成 (mRAG) [7][8]。在该范式下，系统不仅需要理解用户的文本与图像输入，还需要从外部图像、文档、表格等多模态知识库中检索并利用证据。然而，多模态检索器的性能往往成为决定系统上限的核心瓶颈：若检索阶段不能召回真正有用的视觉证据，后续生成模型再强，也容易受噪声图像误导。

MRAG-Bench 正是在这一背景下提出的 vision-centric benchmark。该数据集强调一类典型场景：文本知识并非不可用，而是“不够有效”或“难以获得”，此时额外的图像证据反而更关键 [9]。因此，从 MRAG-Bench 出发研究检索器，不仅有助于回答“图像检索是否有效”，更能进一步回答“检索器的收益如何衡量”与“如何提升”等关键问题，还是下游模型的证据利用能力”。这也是本文工作的核心研究目标。



过程记录

服务部署节点	名称	IP地址	端口	连接	状态
服务节点	服务器	192.168.1.1	8080	成功	运行中
服务节点	服务器	192.168.1.2	8080	成功	运行中
服务节点	服务器	192.168.1.3	8080	成功	运行中

名称	服务时间	服务地址及方式	备注
服务名称	2024-03-15	192.168.1.1:8080	成功
服务名称	2024-03-15	192.168.1.2:8080	成功
服务名称	2024-03-15	192.168.1.3:8080	成功



周记与论文翻译截图。

实验与代码证明

```

1 #!/usr/bin/env python3
2 # -*- coding: utf-8 -*-
3 # @Time : 2024-03-15 10:00:00
4 # @Author : [Your Name]
5 # @Email : [Your Email]
6 # @File : test.py
7
8 import sys
9 import os
10 import time
11 import random
12 import json
13 import requests
14
15 # 配置信息
16 config = {
17     'url': 'http://192.168.1.1:8080',
18     'method': 'POST',
19     'data': {}
20 }
21
22 # 发送请求
23 def send_request():
24     response = requests.post(config['url'], data=config['data'], headers={'Content-Type': 'application/json'})
25     return response
26
27 # 获取响应数据
28 def get_response_data():
29     response = send_request()
30     return response.json()
31
32 # 测试函数
33 def test():
34     for i in range(10):
35         data = {'key': random.randint(1, 1000)}
36         response = send_request()
37         result = get_response_data()
38         print(f"Test {i+1}: {result}")
39
40 # 主函数
41 if __name__ == '__main__':
42     test()
  
```

实验运行、评分和结果流程。

论文内容总览



当前论文初稿的整体排版与内容覆盖情况。

- 1. 研究概述
 - 1.1. 研究目的
 - 1.2. 意义与现状
- 2. 研究内容与方法
 - 2.1. 研究内容
 - 2.2. 技术路线
- 3. 研究进度与结果
 - 3.1. 研究进度
 - 3.2. 阶段结果
- 4. 后续计划与预期效果
 - 4.1. 实验计划
 - 4.2. 预期效果
- 5. 支撑材料
 - 5.1. 论文与过程材料
 - 5.2. 实验与现场展示
- 6. 参考文献

- [1] Lewis, P. et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS, 2020.
- [2] Gao, Y. et al. Retrieval-Augmented Generation for Large Language Models: A Survey. arXiv:2312.10997, 2023.
- [3] Mei, L. et al. A Survey of Multimodal Retrieval-Augmented Generation. arXiv:2504.08748, 2025.
- [4] Abotorabi, M. M. et al. Ask in Any Modality: A Comprehensive Survey on Multimodal Retrieval-Augmented Generation. arXiv:2502.08826, 2025.
- [5] Radford, A. et al. Learning Transferable Visual Models From Natural Language Supervision. ICML, 2021.
- [6] Xie, C. et al. FG-CLIP: Fine-Grained Visual and Textual Alignment. arXiv:2505.05071, 2025.
- [7] Zhang, B. et al. Long-CLIP: Unlocking the Long-Text Capability of CLIP. arXiv:2403.15378, 2024.
- [8] Zhang, K. et al. MagicLens: Self-Supervised Image Retrieval with Open-Ended Instructions. arXiv:2403.19651, 2024.
- [9] Hu, W. et al. MRAG-Bench: Vision-Centric Evaluation for Retrieval-Augmented Multimodal Models. arXiv:2410.08182, 2024.